Adaptive Semantic Memory: Strong Project-State Results, Mixed Generalization to Long-Term Personal Memory

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Preliminary study. c3 is an internal closed set built by the author; EXP-002 uses a public benchmark subset. Not independently reproduced. The ASM implementation is not part of this report. `python3 eval/score.py` reproduces c3 per-method accuracies and paired differences, sign-flip CIs/p-values and scenario counts; LongMemEval subset accuracies, Wilson CIs, mean context estimates, mean per-question evidence recall, paired bootstrap CIs, McNemar p-values, multi-session and abstention counts. c3 marginal bootstrap CIs are independently reproducible from the released scenario counts: use one shared Python RNG seeded 1001, 20,000 draws per method in order ASM / full history / BM25 / summary / diagnostic, sampling 12 scenarios with replacement per draw in sorted scenario order; sort the resulting accuracies and take percentile indices 499 and 19499 (zero-based).

The following are reported from private run logs and are not reproducible from released data: c3 context-token sizes and the inputs to the 12.6x ratio, c3 per-type counts, the decision subset, extraction diagnostics, accounting/retry exclusion and costs.

EXP-001F - internal closed set c3 (built by the author; 216 questions per method, same Qwen3-235B reader):

Method	Accuracy	Mean context, tokens
ASM v4.2	**85.6%**	**489**
Full history	72.7%	6,164
BM25	56.5%	294
LLM summary	42.1%	472

EXP-002 - LongMemEval ORACLE, 100-question stratified subset (public benchmark; not the official 500-question result; GPT-4o judge):

Method	Accuracy	Mean context, tokens
Full history	**66%**	7,683
BM25	58%	995
ASM v4.2	49%	949
LLM summary	49%	305

Strong on project/agent state (internal set), weaker than BM25 and full history on long-term personal memory (public subset). Details, confidence intervals and limitations below.

The problem

In long conversations an agent loses track of what is true *now*: which facts were updated or refuted, which decisions were cancelled or replaced, which goals and constraints still apply, and why things happened. The usual workaround is to re-send the whole history on every call. That gets more expensive as the conversation grows and does not remove the errors: in these tests the full-history reader still repeats stale values and follows instructions planted inside quoted documents.

What ASM does (high level only)

ASM keeps a compact, source-linked record of the current state of a conversation (facts, goals, constraints and decisions, each with a status and the turn it came from). For each question the reader model gets only the relevant part of this state within a fixed token budget (512 tokens in EXP-001F, 1,024 in EXP-002) instead of the raw history. Quoted or forwarded material is kept apart from the user's own goals and rules. Implementation details (memory format, extraction, revision logic) are intentionally not part of this release.

Result 1 - EXP-001F: internal closed set c3 (positive)

• **Internal closed set c3**, built by the author: 12 synthetic long-dialogue scenarios, 18 questions each = **216 questions per method**. Generated after the method was frozen; checked for no overlap with earlier sets. Success criteria pre-registered in git before the first paid call.

• Same reader for all methods: Qwen3-235B-A22B (`qwen/qwen3-235b-a22b-2507`, reasoning off), temperature 0, strict JSON output. Paired comparison on identical questions.

Method	Accuracy (216 q)	95% CI, scenario-clustered bootstrap	Mean context, tokens
ASM v4.2	**85.6%**	84.3-87.0	**489**
Full history	72.7%	68.5-76.9	6,164
BM25 retrieval	56.5%	53.2-59.7	294
LLM summary	42.1%	34.7-49.1	472

ASM used about **12.6x less context** than full history (489 vs 6,164 tokens), as reported from private run logs; these context measurements are not in the release.

Diagnostic condition: the same reader given ASM-style context built from a hand-made reference memory (no automatic extraction) scored 84.7% (CI 82.4-87.0), 0.9 pp below the full ASM pipeline, so on this set extraction was not the accuracy bottleneck.

Paired differences, ASM minus baseline; exact 95% CI from a sign-flip permutation test over the 12 scenarios (all 4,096 sign patterns):

ASM v4.2 vs	Difference	Exact 95% CI	p (exact, two-sided)	Scenarios better / tied / worse
Full history	**+13.0 pp**	+8.4 to +17.8	0.00098	11 / 1 / 0
BM25	**+29.2 pp**	+25.6 to +33.3	0.00049	12 / 0 / 0
LLM summary	**+43.5 pp**	+34.3 to +52.8	0.00049	12 / 0 / 0

Per question type (correct / n), selected (private run logs; counts not released):

Question type	ASM v4.2	Full history	BM25	LLM summary
Planted instruction inside a quoted document	**11/12**	0/12	1/12	0/12
Multi-step cause	24/24	21/24	0/24	2/24
Decision lifecycle	24/24	12/24	12/24	9/24
Look-alike entity names	12/12	10/12	3/12	7/12
Superseded decision	10/12	12/12	11/12	4/12
Revising a conclusion after an upstream fact changes	**0/12**	6/12	0/12	0/12
Telling an intention from a decision	**0/12**	2/12	0/12	0/12

Sources: `docs/exp001f-report.md`, `results/exp001f/final-analysis.json` (accuracy, CI, context, per-type), `docs/exp001f-audit.md` (exact paired CIs) in the private project repository.

Result 2 - EXP-002: LongMemEval ORACLE, 100-question stratified subset (negative)

Frozen ASM v4.2 on a **100-question stratified subset of the LongMemEval ORACLE variant** (cleaned release; stratified by question type and abstention, seed 2002). **This is not the official 500-question result.** Same Qwen3-235B reader for all methods; answers graded with the official LongMemEval GPT-4o judge (`gpt-4o-2024-08-06`, upstream prompts), blind to method. Budget 1,024 tokens (characters / 4) for the compressed methods.

Method	Accuracy (100 q)	95% CI (Wilson)	Mean context, tokens (chars / 4)	Evidence recall
Full history	**66%**	56.3-74.5	7,683	100%
BM25 retrieval	58%	48.2-67.2	995	77.5%
ASM v4.2	49%	39.4-58.7	949	40.7%
LLM summary	49%	39.4-58.7	305	0%*

Evidence recall is mean per-question evidence recall over 95 questions per method with non-null recall fractions. It is not the percentage of questions with any evidence present. *The summary rewrites text, so literal-turn recall is zero by construction; this does not measure semantic coverage.

Paired, ASM minus other (stratified bootstrap, 10,000 repeats; exact McNemar):

Comparison	Difference	95% CI	McNemar p
ASM - BM25	-9 pp	-17 to -1	0.064 (borderline, not significant at 0.05)
ASM - full history	-17 pp	-26 to -8	0.0015
ASM - LLM summary	0 pp	-10 to +10	1.0

ASM scored 6/27 on multi-session questions, versus 11/27 for BM25 and 17/27 for full history. Its lowest observed type accuracy was single-session-preference (1/6). It abstained correctly on 5 of 6 unanswerable questions (too few for a claim). Mean per-question evidence recall was 40.7% for ASM versus 77.5% for BM25, consistent with evidence selection being a limitation.

Source: `docs/exp002-oracle-report.md`, `results/exp002o/final-analysis.json` (EXP-002 worktree of the private repository).

The two experiments use different task sets and are not directly comparable. Together they suggest ASM is promising for agent / project state tracking (cancelled and replaced decisions, provenance, injected instructions), but it is **not** a general personal-memory system yet.

Limitations (read before citing)

• **Internal benchmark.** EXP-001F uses the internal closed set c3: synthetic, non-English (mixed Ukrainian/Russian) dialogues that I generated, one reader model, and has not been reproduced externally.

• **External benchmark: negative.** On the LongMemEval ORACLE 100-question subset ASM trails BM25 and full history. The S (long-history) variant has not been tested.

• **Decisions not clearly better.** On the 72 decision-related questions of the internal closed set c3 ASM is +2.8 pp over full history, 95% CI -2.8 to +8.3 (includes zero).

• **Failing question types.** Revising a conclusion after an upstream fact changes (0/12) and telling an intention from a decision (0/12).

• **False updates.** The memory-update step's false-update rate on the internal closed set c3 is 20.5% (pre-registered target <= 10%); its recall of decision accept / cancel / replace events is 93.8%.

• **Accounting flag.** 12 API calls in the internal closed set c3 run timed out and were retried. An offline audit confirmed all 216 questions of every method were answered and graded once and the spend was recorded conservatively. Under the strict pre-registered accounting rule the run remains formally flagged `valid: false`; I have not changed that flag. Excluding the 3 affected reader questions gives ASM 85.4% vs full history 72.3% (audit).

• **Small samples.** 12 scenarios (internal closed set c3) and 100 questions (EXP-002): only differences of roughly 10 pp are detectable on EXP-002.

• **Cost of ASM per question is higher** than full history on the internal closed set c3 at this scale because of the memory-update calls (about $0.00099 vs $0.00046 per question, `exp001f-report.md`), even though the reader context is ~12.6x smaller.

Cost and process

• All paid experiments to date cost about **$1.87** in total (OpenRouter key counter), including **$0.44** for the internal closed set c3 run (key delta) and **$0.42** for EXP-002.

• Pre-registered criteria, frozen configuration with SHA-256 manifest, replayable response cache. Experiment code reviewed by a second AI system before each paid run.